

David Rosocha

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EXPERIENCE

- **Ciena** Ottawa, Canada
FPGA Design Co-op Jan 2026 – Apr 2026
 - Built a hardware tracer core in SystemVerilog capturing and timestamping up to 256 live signals at sub-10ns resolution, with triggers on signals, vectors, and peripheral protocols (AXI, I2C), replacing commercial ChipScope tooling
 - Implemented AXI-Streaming and MPI interfaces for DMA transfer between the tracer core and an external CPU, enabling real-time signal analysis on Linux workstations*FPGA Verification Co-op* May 2026 – Aug 2026
 - Developed UVVM verification environments for the FPGA logic behind Hyper Rail, Ciena's flagship inter-data-center interconnect, verifying functional correctness, interface compliance, and timing
 - Built constrained-random stimulus and protocol-level agents to drive coverage closure across core IP blocks, catching functional and timing bugs pre-silicon
- **SHaRe Lab, Carleton University** Ottawa, Canada
ML Hardware Researcher, supervised by Dr. Arash Ahmadi Sept 2026 – Present
 - Researching whether the event-driven sparsity of **spiking neural networks** yields measurable energy savings on FPGA, extending the published [Spiking Music](#) event-based audio compression algorithm to hardware
 - Leading the hardware implementation: fixed-point software model to synthesizable SystemVerilog with bit-exact verification, using on-board power and latency as the research metrics; targeting a conference submission
- **National Research Council Canada** Ottawa, Canada
Integrated Photonics Researcher, supervised by Dr. Ross Cheriton Sept 2026 – Present
 - Researching and designing silicon nitride photonic devices for **on-chip gas sensing** (Lumerical, KLayout)
- **Telesat** Ottawa, Canada
LEO Satellite Bus & Systems Engineering Co-op May 2025 – Aug 2025
 - Designed a satellite coverage simulation tool in Python (NumPy/pandas), achieving a **300% runtime improvement** for 1400+ LEO satellites and replacing commercial tooling (e.g., STK)
 - Built an Optical Inter-Satellite Link framework simulating azimuth, Doppler shift, and alignment across **entire constellation orbit configurations**, informing crosslink feasibility
- **uODD** Ottawa, Canada
Founder Dec 2024 – Present
 - Founded the first digital design club at uOttawa, growing to **160+ members in 8 months**; coordinated FPGA workshops and authored open-source Verilog curriculum with **Ciena and Synopsys**

KEY PROJECTS

- **FPGA Transformer Accelerator (TPU)** [GitHub](#) — [Video](#)
SystemVerilog/UVVM/Vivado/Quarta/PyTorch 2026
 - Designed and verified a TPU-style transformer inference accelerator on a Basys 3 (Artix-7) FPGA, offloading the attention layers of a trained neural network to hardware over a UART host interface
 - Implemented an **8×8 output-stationary systolic array** (64 MAC processing elements) with a double-buffered tiling controller scaling arbitrary matrix multiplies up to 64×64, sized to the board's 90 DSP slices
 - Designed a divider-free **hardware softmax** using a 2D-LUT approximation, validated to <0.9% mean error against a floating-point reference in Q8.8 fixed-point
 - Built a **UVM environment** from scratch (agent, driver, monitor, scoreboard, constrained-random stimulus, coverage), checking the datapath against a golden model to catch silent fixed-point errors
 - Scripted, animated, and edited a technical demo video explaining the architecture, reaching **100k views** on YouTube

TECHNICAL SKILLS

- **Hardware:** SystemVerilog, Verilog, VHDL, RTL/FSM design, fixed-point, AXI/AXI-Stream, I2C, UART, timing closure
- **Verification:** UVM, UVVM, SVA, constrained-random stimulus, functional coverage, scoreboards, golden models
- **Tools:** Vivado, Questa, ModelSim, Quartus, KLayout, Lumerical, Git, Linux, TCL, Python (PyTorch), C/C++

EDUCATION

- **University of Ottawa** Ottawa, Canada
BASc Electrical Engineering & BSc Computing Technology (Double Degree) Expected Dec. 2027
 - **GPA: 3.99/4.00** (CGPA: 9.88/10.0) — **Faculty of Engineering Co-op Student of the Year 2025**